USM Orbit

Service & Maintenance Operations Platform

Project Plan & Status

One Platform. Every Service & Maintenance Operation.

A consolidated record of everything planned and built so far — from the USM Orbit business plan, through the market analysis and product specification, to the working beta application, its current status and the next steps.

Version 1.2 · 28 October 2026 Status: Beta built · ready for test hosting Confidential — internal planning

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Sources consolidated in this document

- *Global Facilities Operations Platform Report* (strategy and market analysis).
- *USM Orbit Beta Build Specification* (23 Oct 2026).
- *USM Orbit – Business Plan*, compiled by Krishna, 19 Oct 2026.
- The working beta application, repository `satya928/USMorbit`, and its documentation.

1 Executive summary

USM Orbit is a cloud, multi-tenant platform that runs service and maintenance operations end to end — from a client's request through assignment, field execution, quality inspection and client sign-off to reporting — with compliance, evidence and an audit trail built in.

Beta built
Working web application with real database, logins and permissions

≈70%
of the business plan's Phase 1 MVP already implemented

56
automated tests passing (51 unit/integration + 5 browser end-to-end)

Where we are

- **Plan defined.** The **USM Orbit** business plan sets out the product, market, revenue model and roadmap; a market analysis sharpened the positioning as an AI-native service and maintenance operations platform.
- **Software built.** A production-minded beta covers Command Center, Work Orders, Sites, Assets, People & Contractors, Compliance, Inspections, Approvals, Reports, AI Copilot and Settings — all backed by PostgreSQL, with tenant isolation enforced in the database itself.
- **Ready for testers.** The code is on GitHub with a one-click free hosting blueprint (Render), a tester guide, nine fictional demo accounts and a feedback channel. It is **not yet hosted** — a hosting account owned by the team is the only remaining step to get a public test link.

Decisions and actions needed now

Item	Recommendation
Brand protection	The name **USM Orbit** is confirmed. Check trademark and domain availability before external launch.
Test hosting	Deploy the Render blueprint (free, ~15 min) for a shareable link; move to a ~$5–10/month VPS for anything longer-lived.
Feedback form	Create a Google Form and set it as the in-app "Beta feedback" link.
AI key	Optional for testing; add an Anthropic API key to switch the Copilot from labelled demo mode to real AI answers.
Next build sprint	Close Phase 1 gaps: site attendance (GPS check-in/out, timesheets), Projects & Scope of Work, client feedback ratings.

2 Origin & timeline

USM Orbit originates from the **USM Orbit business plan** (Universal Service & Maintenance Orbit), compiled by Krishna on 19 October 2026. Everything that followed — the market analysis, the product specification and the beta application — was built to deliver that plan.



The operational core of USM Orbit

Client → Contract → Work → Worker → Supervisor → Quality → Compliance → Sign-off → Reporting

1 — The USM Orbit business plan

The plan defines USM Orbit as a configurable SaaS platform combining **Field Service Management** and **Facility Maintenance Management**: eleven modules (workforce, work orders, field operations, work execution, quality, client portal, compliance, safety, inventory, Orbit Academy, contractor management), subscription pricing and a five-phase product roadmap, with commercial cleaning and facility maintenance as the first market.

2 — Strategy & market analysis

The analysis mapped the global facilities-software landscape and concluded that USM Orbit should position itself not as "another facility management software" but as an **AI-native operating system for service and maintenance operations** — coordinating people, sites, assets, contractors, work, compliance and clients — starting with outsourced facility services.

3 — Beta build specification

A detailed specification turned the plan's Phase 1 priorities into a buildable beta: modules, roles and permissions, database design, workflow rules, security requirements, KPI definitions, AI guardrails and acceptance tests.

4 — Beta application

The specification was implemented as a working, multi-tenant web application with real logins, persistent data, automated tests, a test-hosting kit and documentation. It is the MVP engine of the USM Orbit plan, branded with the plan's name and tagline.

3 Vision, positioning & business model

Vision

To become the leading global Service Operations Platform that enables organisations to manage every aspect of service delivery from a single connected ecosystem.

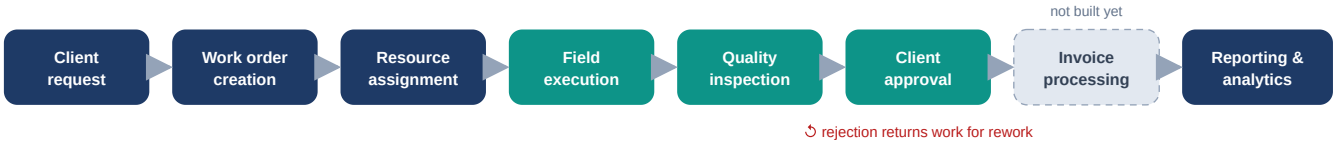
Mission

To simplify, standardise and digitalise service and maintenance operations through configurable technology that improves productivity, compliance, quality and customer experience.

The problem

Service businesses still run on disconnected tools — spreadsheets, paper reports, email, shared folders, standalone scheduling, compliance and inventory systems. The result: communication delays, weak accountability, inconsistent quality, compliance risk, slow invoicing and high admin cost.

The solution — one operational lifecycle



Value proposition

Audience	Value
Business owners	Productivity, profitability, visibility, reduced operational risk
Operations managers	Real-time workforce visibility, better scheduling, faster issue resolution
Field teams	Mobile-first experience, clear instructions, simple reporting
Clients	Transparency, faster communication, digital sign-off, quality assurance

Target markets

Primary

Facility maintenance companies — commercial cleaning, strata, property and grounds maintenance, building services.

Secondary

Trades (electrical, plumbing, HVAC, fire, security), **construction** (fit-out, upgrades), **asset management** (building and property managers).

Future verticals

Healthcare (hospitals, aged care), education, government and councils, industrial (manufacturing, distribution).

Business model

Stream	Detail
Subscription plans	**Starter** (small service businesses) · **Professional** (growing organisations) · **Enterprise** (multi-location)
Per-user licensing	Scales with workforce size. The strategy analysis also suggested hybrid per-site pricing and premium AI / IoT / vision modules.
Implementation services	System setup, configuration, data migration
Training services	Administrator and user training
Premium support	Dedicated account manager, SLA support
API & integrations	Custom integrations (ERP, HR, accounting, building systems)

Go-to-market: start with commercial cleaning and facility maintenance (validated workflows), via direct sales to medium-sized service businesses, partnerships with industry associations and maintenance networks, digital marketing (SEO, LinkedIn, webinars) and referrals.

4 Market & competition

Facility-management software is a large, growing and competitive category. The strategy analysis cited an estimated market of about **US$4.1 billion in 2025, forecast to roughly US$9.6 billion by 2033** (estimates vary by research definition). The industry is moving toward mobile execution, IoT, predictive maintenance, vendor management, spatial intelligence and — most recently — embedded AI agents.

Competitive layers

Layer	Examples	Opportunity for Orbit
CMMS / maintenance	MaintainX, UpKeep, Limble, Fiix	Add contractor, client and compliance layers they lack
CAFM / FM operations	ServiceChannel, FMX, Facilio	Operations-first, simpler, outsourced-services focus
IWMS / enterprise FM	Planon, Eptura, IBM, Archibus	Faster to deploy; avoid real-estate/lease complexity
Building / IoT / BMS	Siemens, Honeywell, Schneider, Johnson Controls	Integrate later rather than compete
Enterprise workflow	ServiceNow	Deeper FM-specific vertical workflows

Key benchmarks

ServiceChannel

The closest benchmark: multi-site service, work orders, invoices, provider search, contractor scorecards, compliance. Reported to have launched an AI operating system for FM in 2026.

Facilio

Connected buildings plus AI; reported to have launched autonomous FM agents (intake, approvals, invoice validation, reporting) in 2026.

MaintainX / UpKeep

Mobile-first CMMS: work requests, PM, procedures, audit history. Strong on technicians, weaker on contractors and clients.

Planon / IBM Maximo

Enterprise IWMS / EAM breadth (real estate, assets, sustainability). Powerful but heavy; not a day-one target.

The white space

Most systems centre on *Asset → Maintenance → Technician*. USM Orbit's core workflow is *Client → Contract → Work → Worker → Supervisor → Quality → Compliance → Sign-off → Invoice*. That makes it a natural **operating system for outsourced facility services** — a focused beachhead from which to expand into broader FM.

What we deliberately do not build first

Lease accounting, full real-estate financial management, CAD/BIM replacement, building-control systems, ERP, HR/payroll, full procurement ERP and complete energy management. Integrate with these instead.

5 Strategic differentiation

Everyone has work orders, mobile apps and, increasingly, a chatbot. Orbit should compete on **facility intelligence and autonomous operations** — continuously answering: *what is happening, why, what happens next, what should we do, and can the system do it?*

1. Facility events, not just tickets

A request becomes an event the system understands: impact, asset, location, vendor, SLA, compliance — then creates work, assigns, monitors, validates, updates asset history and learns.

2. Facility knowledge graph

Relationships across building → floor → room → asset → maintenance → technician → contractor → contract → SLA → invoice, so AI can explain *why* costs or failures rise.

3. AI that operates, not chats

Specialised agents: work-order triage, dispatch, compliance, contractor, asset, inspection, invoice, safety, reporting and knowledge agents — with permissions and human approval.

4. Vision-based inspections

Photos analysed for leaks, damage, blocked exits, PPE and cleanliness issues → severity → work order → verification.

5. Compliance that answers "now"

Is this site, worker, contractor, asset and job compliant right now? Proactive alerts and assignment blocks — already started in the beta.

6. Marketplace & network effects

Long term: a vendor marketplace (request → quote → award → work → inspection → invoice → rating) feeding a data flywheel.

The moat

Not the UI, but: the facility knowledge graph, the operational dataset, AI decision models, the vendor performance network, compliance knowledge, workflow templates, integrations, customer history and agent execution infrastructure.

Five AI products over time

Product	Opportunity	Status
FM Copilot	Ask anything about facilities, grounded in records	Started in beta
FM Autopilot	Agents execute routine work with approvals	Future
FM Vision	AI visual inspection of photos	Future
FM Predict	Predict failures and costs	Future (needs data)
FM Intelligence	Portfolio-level optimisation	Future

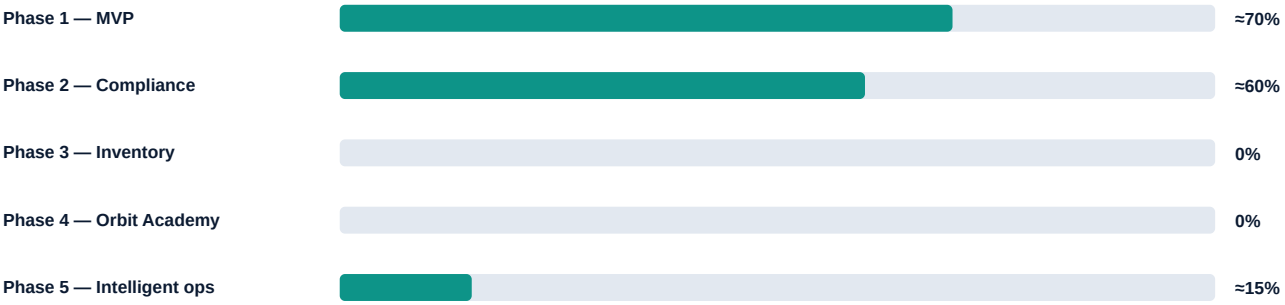
6 Product scope & module status

The business plan defines eleven modules. The table compares each with the beta that has been built.

Built working in the beta Partial core works, some features missing Not built planned Ahead of plan built earlier than planned

Module (plan)	Status	Built in the beta	Still to build
Workforce Management	Built	Employee & contractor profiles, skills, licences and certifications with expiry	Availability calendar, onboarding workflows
Project & Work Orders	Partial	Full work-order lifecycle, assignment, progress, status updates, SLA	Projects, Scope of Work, task delegation within projects
Field Operations	Not built	Activity logging (timeline)	GPS attendance, mobile check-in/out, timesheets, site access
Work Execution	Built	Digital checklists, photo evidence, completion and progress tracking	—
Quality Assurance	Built	Inspection templates, audits, scores, defects, corrective work orders, photo documentation	KPI rating systems beyond scores
Client Experience Portal	Partial	Client approver role: sign-off, reject/rework, service requests, completion verification	Feedback & ratings, branded portal, inspection walkthroughs
Compliance Management	Partial	Insurance, licence and certificate tracking, verification, documents, 30/60/90-day alerts, assignment blocking	SWMS/SWI library, policy distribution with acknowledgement
Safety Management	Not built	—	Toolbox talks, site inductions, safety alerts, WHS comms, incident reporting
Inventory Management	Not built	—	Stock audits, site/vehicle inventory, reorder, procurement
Orbit Academy	Not built	—	Training content, videos, knowledge base, certification learning
Reporting & analytics	Built	Command Center KPIs and charts; 8 reports with CSV export	Advanced/custom dashboards
AI features	Ahead	AI Copilot with read-only tools and draft-then-confirm work orders	Predictive maintenance, scheduling, AI quality reviews
Invoice processing	Not built	Estimated and actual costs per job	Invoices linked to approved work

Coverage at a glance



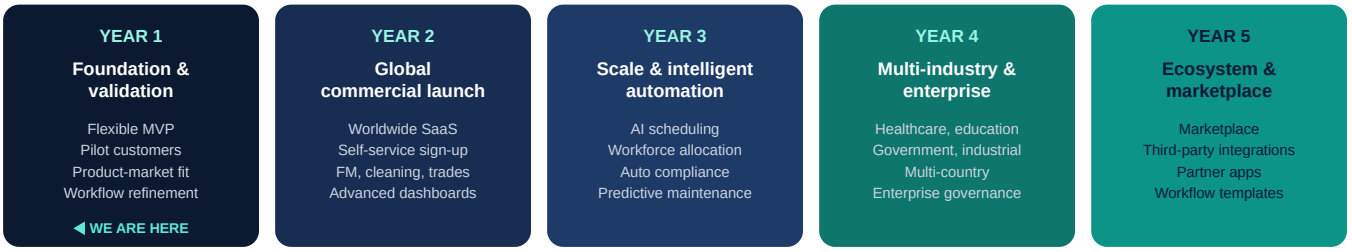
Percentages are qualitative estimates of feature coverage against each phase's listed items, not effort estimates.

7 Roadmap

Product roadmap (business plan phases)

Phase	Scope	Status
1 — MVP: core operations	Employee management, work orders, project management, site attendance, quality audits, client sign-off	≈70% Missing: project management, site attendance
2 — Compliance	Licences, certifications, insurance, policies, SWMS, SWIs	≈60% Missing: policies, SWMS/SWI library
3 — Inventory	Inventory tracking, stock audits, purchase requests, procurement	Not started
4 — Orbit Academy	Learning management, video training, certification pathways	Not started
5 — Intelligent operations	Predictive maintenance, workforce optimisation, automated scheduling, AI quality reviews, performance insights	Started early AI Copilot

Strategic growth roadmap (years 1–5)



Recommended near-term build sequence

1. **Test round 1 (now):** host the beta, run pilot testing with 3–5 friendly users from cleaning / facility maintenance, collect feedback.
2. **Sprint A — finish Phase 1:** site attendance (mobile check-in/out with GPS, timesheets), Projects & Scope of Work grouping work orders, client feedback & ratings.
3. **Sprint B — finish Phase 2:** SWMS/SWI and policy library with distribution and acknowledgement; safety basics (toolbox talks, inductions, incident reports).
4. **Sprint C — commercial readiness:** email notifications and password reset, invoicing from approved work, S3-compatible file storage, Microsoft / SAML SSO for enterprise pilots.
5. **Then:** Phase 3 inventory, Phase 4 Academy, and deeper AI (scheduling, vision on inspection photos) once real data exists.

8 The beta application

The beta is a working, database-backed web application — not a mockup. It was built from the *Beast Mode* specification, prioritising one complete, reliable workflow over feature count.

Core workflow

Request → Triage → Assignment → Work in progress → Inspection / evidence → Client approval or rework → Closure → Reporting. Every transition is validated on the server and recorded with the actor, time and reason.



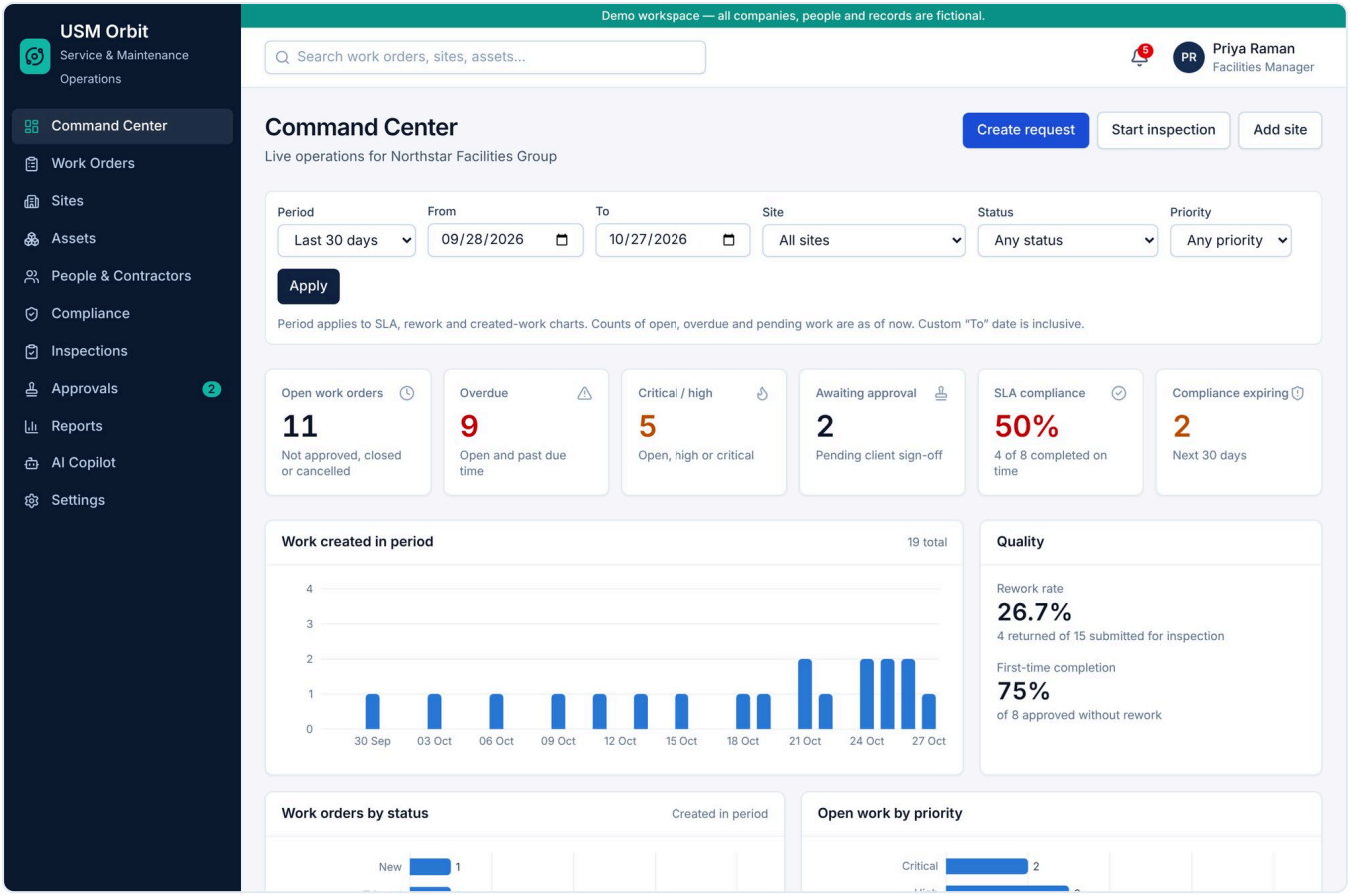
Roles and permissions

Role	What they can do
Tenant Admin	Everything in the workspace: users, roles, site access, settings, audit log
Facilities Manager	Create/assign work, approve, close, sites, assets, compliance, inspections, reports, AI Copilot
Supervisor	Manage execution and inspections on authorised sites; approve only if granted
Technician / Cleaner	Only their own jobs: start, checklist, photos, comments, submit for inspection
Contractor Manager	Only their company's jobs, staff and compliance
Client Approver	Approve / reject / request rework on authorised sites; raise requests; never sees internal notes
Read-only Auditor	View everything in scope, audit log, report exports; no edits

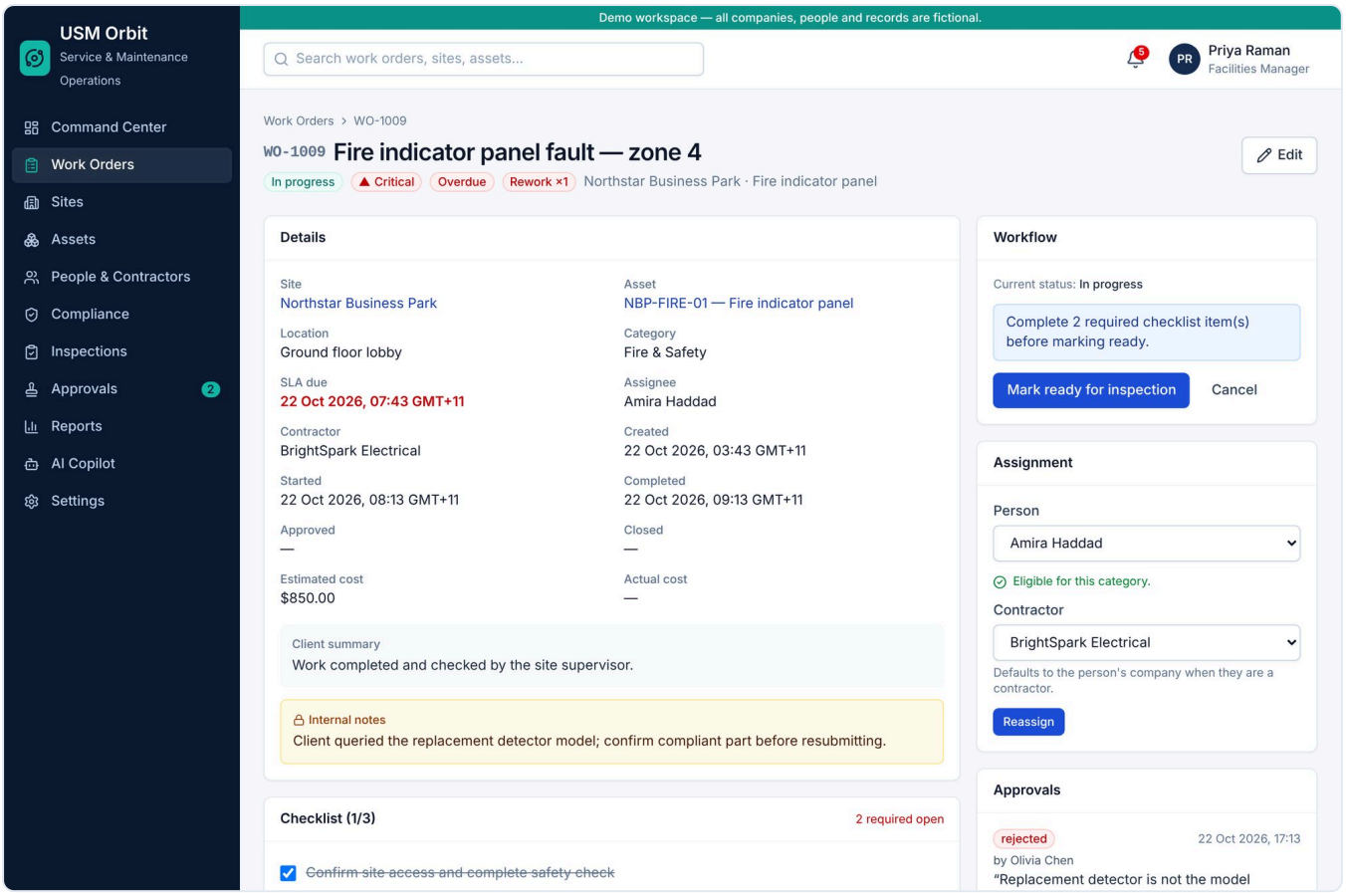
Modules delivered

Command Center — live KPIs, charts, filters, site performance, compliance expiries, activity feed.
Work Orders — search/filter, create/edit, assignment, checklist, evidence, comments, timeline.
Sites & Assets — portfolio, asset register, history, rule-based health indicator.
People & Contractors — profiles, credentials, contractor performance.
Compliance — expiry views, verification, documents, alerts, assignment rule.
Inspections — templates, runs, defects → work orders, sign-off.
Approvals — client and inspection queues.
Reports — 8 reports + CSV. **AI Copilot** — grounded answers, drafts. **Settings** — users, roles, SLA, categories, audit.

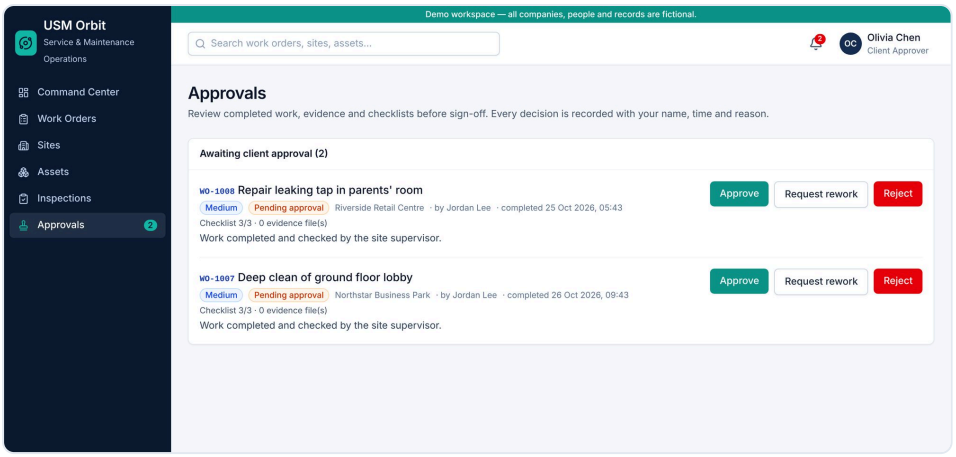
Screens



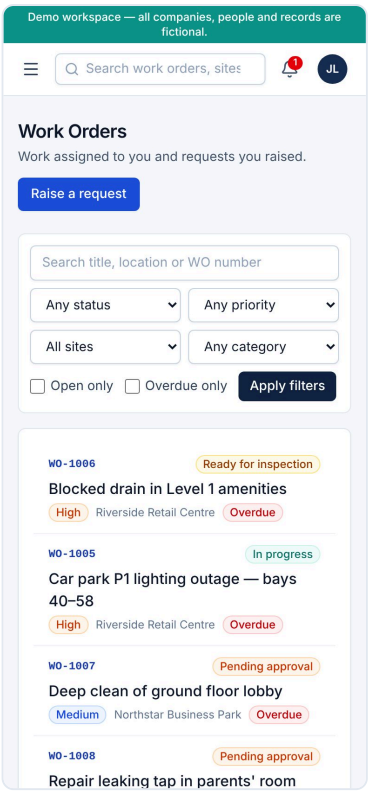
Command Center — every figure is calculated from database records; filters apply to all KPIs and charts.



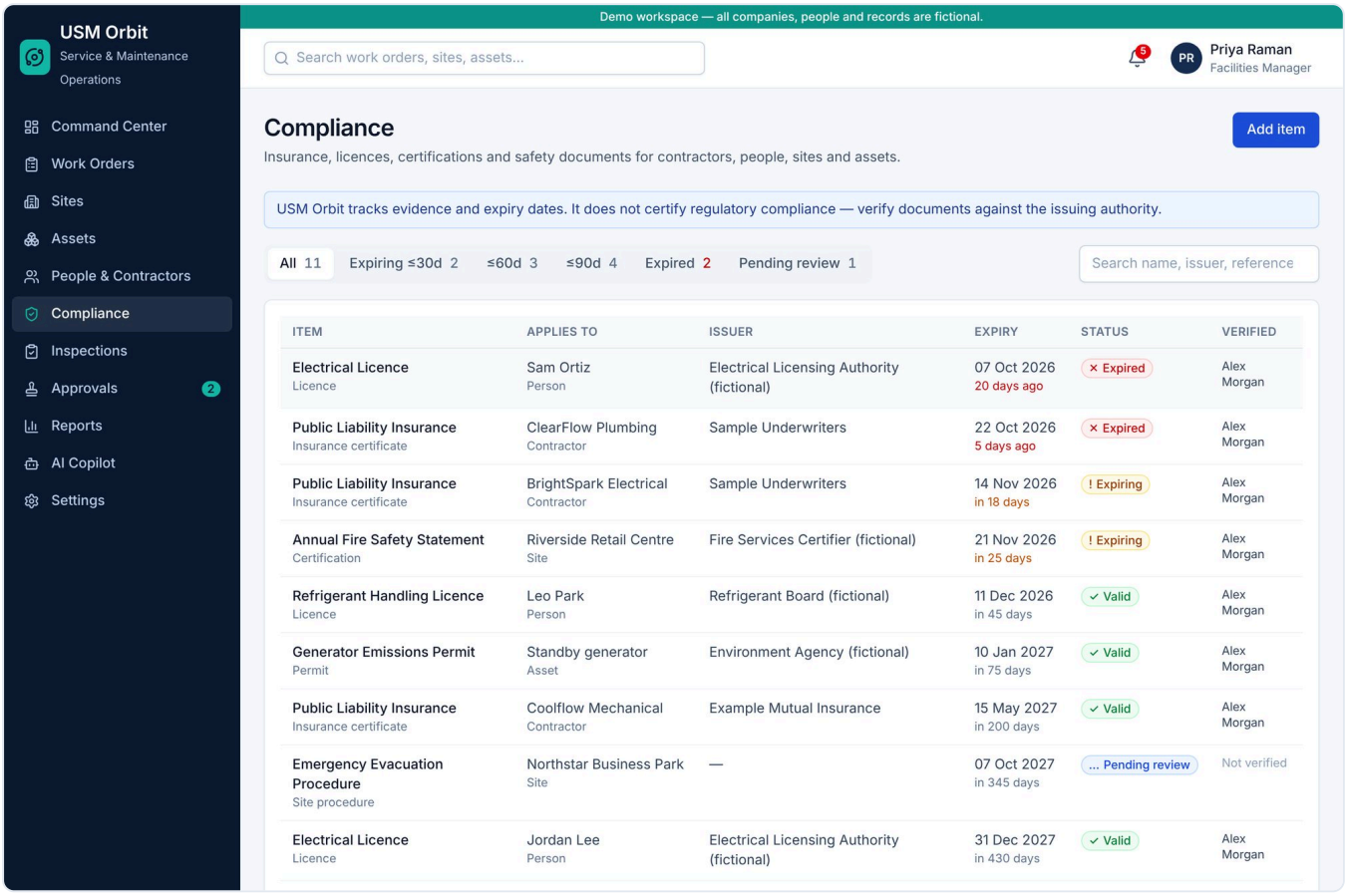
Work-order detail — workflow actions for the signed-in role, assignment with credential check, checklist, evidence, internal notes, approvals and a full timeline (including the client's rejection reason).



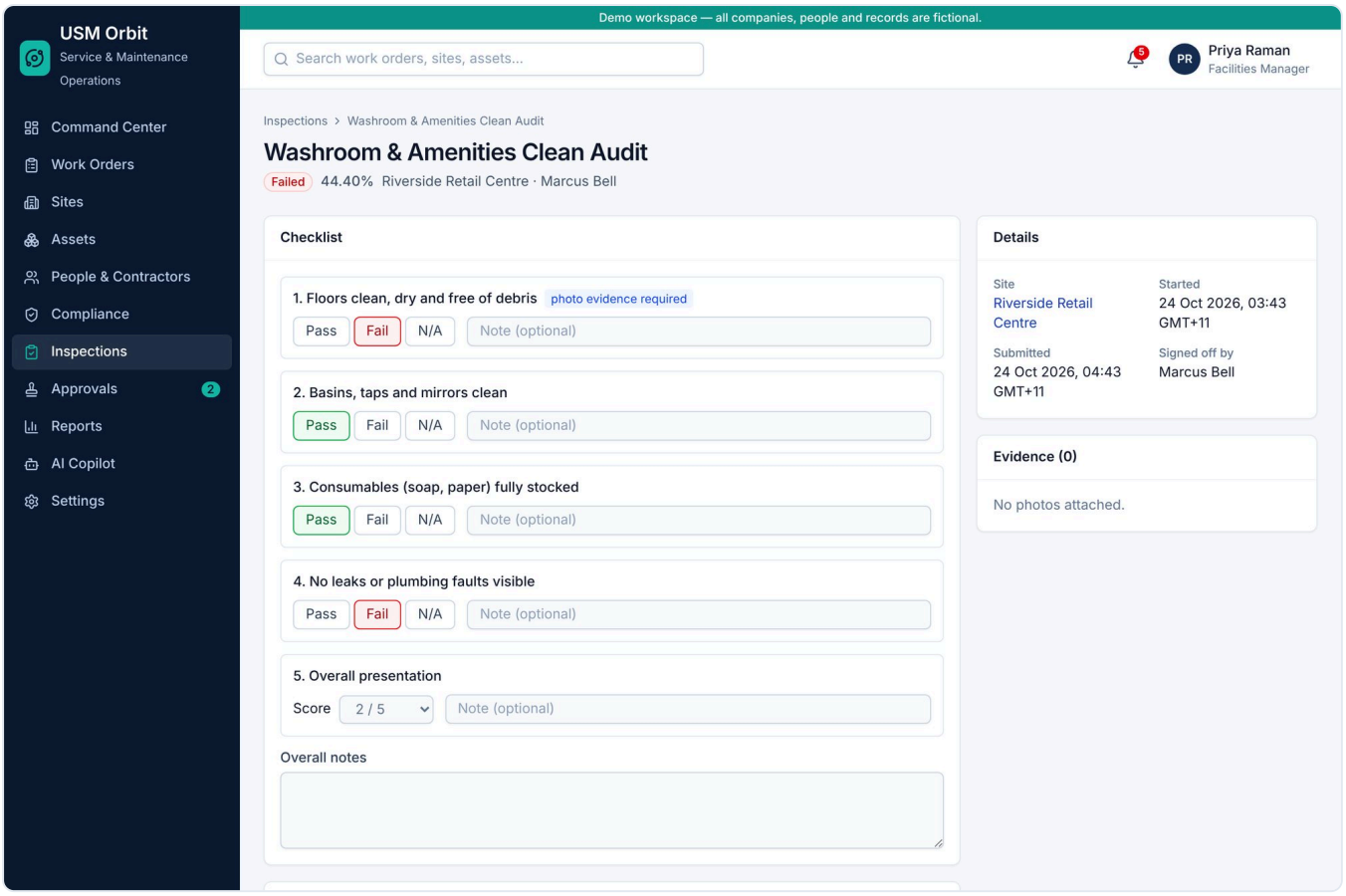
Client approval queue — approve, request rework or reject with a reason; self-approval is blocked.



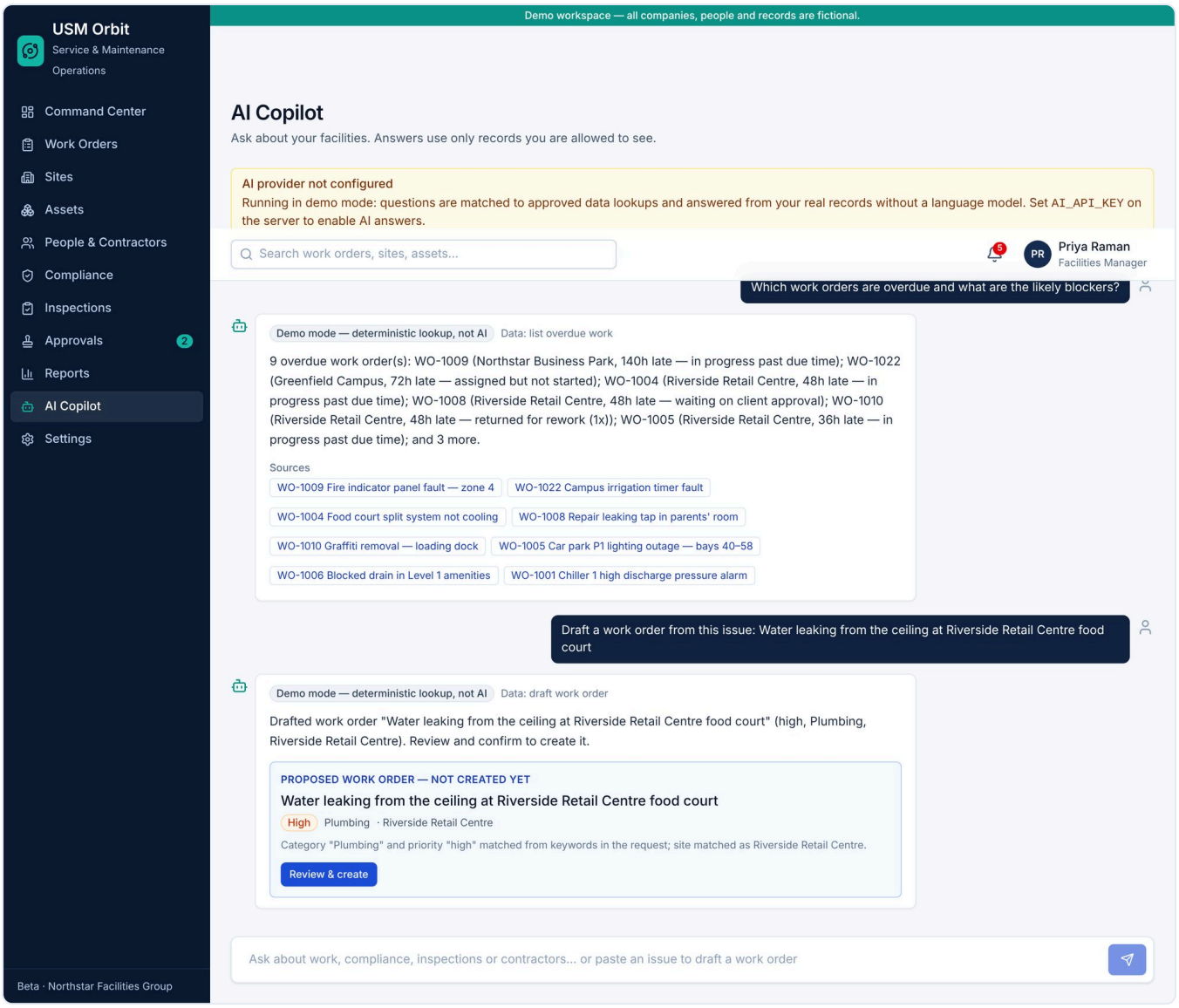
Technician view on a phone — only their own jobs.



Compliance register — 30/60/90-day and expired views, verification status; a tracking aid, not a certification.



Inspection run — pass/fail/score answers, defects with severity, one-click work order from a defect, photo evidence.



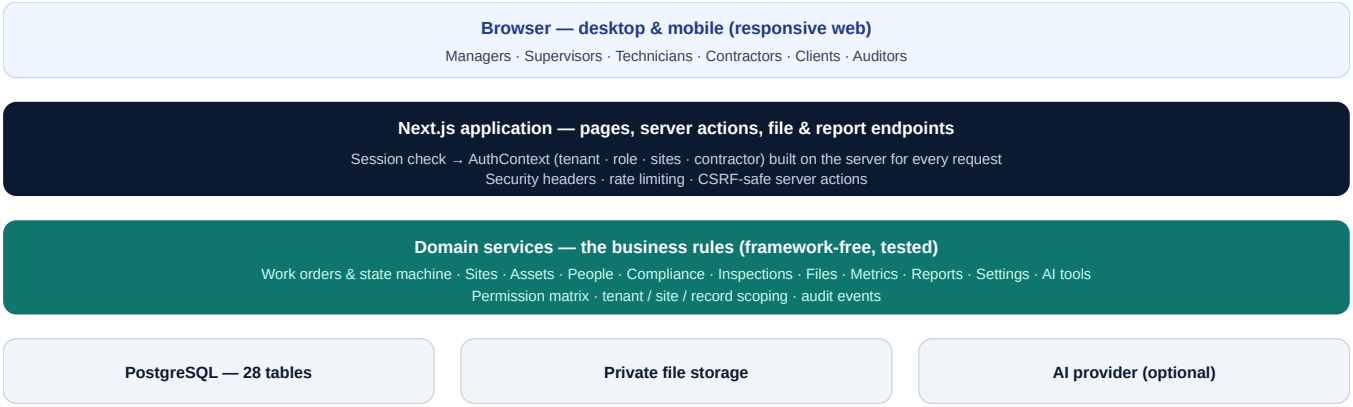
AI Copilot — answers from authorised records with source links; drafted work orders are only created after the user reviews them. Shown here in labelled demo mode (no AI key configured).

9 Technology, security & quality

Stack

Layer	Technology	Why
Application	Next.js 15, React 19, TypeScript	One deployable unit: screens, server logic and APIs
Design	Tailwind CSS 4, Recharts, lucide icons	Responsive, accessible UI on desktop and phone
Database	PostgreSQL 16 + Drizzle ORM	Reliable relational data, versioned SQL migrations
Authentication	Email/password with database sessions + optional **Sign in with Google**	Self-hostable; Google sign-in for invited users only
Validation	Zod	Every input checked on the server
Files	Private storage + permission-checked downloads	Evidence and documents never publicly exposed
AI	Claude via the official Anthropic SDK (default model `claude-opus-5-5`)	Provider adapter; labelled demo mode without a key
Testing	Vitest + Playwright	Unit, integration (real database) and browser tests
Hosting	Docker, Docker Compose, Caddy (HTTPS); Render blueprint	VPS or free cloud test hosting

Architecture (modular monolith)



Security controls

- Hashed passwords (scrypt); session tokens stored only as hashes; secure cookies
- Optional Sign in with Google (OpenID Connect + PKCE) — invited users only, linking audited
- Server-side role permissions; site and record scoping on every query
- **Tenant isolation enforced in the database** (composite foreign keys)
- Validated workflow transitions; self-approval blocked; append-only timeline
- File type checks by content; private storage; access re-checked on download
- AI is read-only, permission-scoped; only usage metadata logged

Quality

- **51** unit & integration tests against a real PostgreSQL database
- **5** browser end-to-end flows (full lifecycle across roles incl. rework, credential override, access control, CSV export & Copilot, Google sign-in)
- Lint, type checks and production build pass
- 36 screens, 3 API endpoints, ~12,900 lines of TypeScript
- Not certified (SOC 2 / ISO 27001 / GDPR) — do not claim compliance

10 Current status, storage & how to test

Item	Status
Code	GitHub repository `satya928/USMorbit`, branch `main` — all work committed and pushed
Live test link	**None yet** — needs deployment to a hosting account owned by the team
Database	Created and filled with fictional demo data automatically on first start of a hosted instance
Documentation	README, tester guide, demo script, deployment guide, feature checklist, security notes, this document
Branding	USM Orbit — logo, tagline, demo password `Orbit-Demo-2026`

Hosting options

Option	Cost	Notes
Render + free Postgres (recommended for testing)	Free	One-click blueprint in the repo. Sleeps after ~15 min idle; free database expires after 30 days; uploads lost on restart.
VPS (Hetzner, DigitalOcean, Hostinger…)	~$5–10 / month	Docker Compose + automatic HTTPS; backups script included. Best for longer pilots.
Laptop + tunnel	Free	Quick demo link; only works while the laptop is on.
Google Cloud Run + Cloud SQL	Trial credits	Google option; needs cloud file storage added.
Google Blogger / Drive	—	**Not possible** — they cannot run server code or a database. Use Drive for sharing documents and videos instead.

Getting the test link (Render, ~15 minutes)

1. Sign up at **dashboard.render.com** (sign in with GitHub) and allow access to `satya928/USMorbit`.
2. **New → Blueprint**, pick `satya928/USMorbit` (branch `main`).
3. Leave `AI_API_KEY` empty (demo mode) or add a key; optionally set `FEEDBACK_URL` to a Google Form. Click **Apply**.
4. After 5–10 minutes Render shows the link, e.g. `https://usm-orbit-beta-xxxx.onrender.com`. Share it with the tester guide.

Demo accounts (fictional data) — password `Orbit-Demo-2026`

Role	Email
Facilities Manager	`manager@northstar.demo`
Technician	`tech@northstar.demo`
Technician (expired licence)	`tech2@northstar.demo`

Role	Email
Supervisor	`supervisor@northstar.demo`
Client Approver	`client@northstar.demo`
Contractor Manager	`contractor@northstar.demo`
Tenant Admin	`admin@northstar.demo`
Read-only Auditor	`auditor@northstar.demo`
Separate company (isolation check)	`admin@harbourline.demo`

What testers check

A. Full job lifecycle across manager → technician → supervisor → client → manager, including a rejection and rework. B. Credential rule (expired licence blocks assignment). C. Inspection with a failed item and a defect → work order. D. Dashboard filters, report CSV export, AI Copilot. E. Permissions (technician restrictions, clients never see internal notes, second company sees nothing of Northstar). F. Technician experience on a phone. Feedback via the in-app button or GitHub issue templates.

11 Gaps, risks & next steps

Known limitations of the beta

- Not yet hosted; free hosting has sleep, 30-day database and file-persistence limits.
- In-app notifications only — no email or password-reset email. Sign in with Google is available; enterprise SAML SSO and SCIM are not yet.
- Responsive web only — no native iOS/Android apps or offline mode yet.
- Single-server assumptions (local file storage, in-memory rate limiting).
- English only; no inventory, attendance, projects, safety, academy or invoicing yet.
- Asset health is rule-based, not predictive; AI answers are demo mode until a key is added.

Risks and mitigations

Risk	Mitigation
Crowded market with well-funded competitors already shipping AI agents	Focused beachhead (outsourced facility services), contractor + client + compliance workflow depth, fast pilots
Building too broadly (11 modules) before product-market fit	Finish Phase 1, validate with pilots, then expand by evidence
Enterprise security expectations	Tenant isolation in place; add SSO, MFA, backups testing and an external security review before paid enterprise pilots
AI accuracy and trust	Read-only AI, human confirmation for every change, clear labelling, verification reminders
Brand conflicts	Check trademark and domain availability for USM Orbit before external launch

Next steps checklist

#	Action	Owner
1	Check trademark and domain availability for USM Orbit	Founders
2	Create Render account and deploy the blueprint; share the link	Team
3	Create a Google Form for beta feedback and set `FEEDBACK_URL`	Team
4	Optional: set up Google sign-in credentials (docs/GOOGLE_SIGN_IN.md)	Team
5	Recruit 3–5 pilot testers from cleaning / facility maintenance	Founders
6	Optional: add an Anthropic API key to enable real AI answers	Team
7	Build Sprint A: site attendance, Projects & SOW, client feedback	Development
8	Build Sprint B: SWMS/SWI & policy library, safety basics	Development

Appendix A — KPI definitions

Implemented once and used identically on the dashboard, in reports and by the AI Copilot.

KPI	Definition
Open work orders	Status not approved, closed or cancelled
Overdue	Open and past the due time
Critical / high	Open with priority high or critical
Awaiting approval	Status pending client approval
SLA compliance	Of approved/closed work with a due time completed in the period, the share completed on or before due (cancelled excluded; "—" if none)
Expiring compliance	Live items expiring within the next 30 days
Rework rate	Returns for rework ÷ submissions for inspection in the period
First-time completion	Of work approved in the period, the share never returned for rework
Compliance status	Not applicable → expired → pending review → expiring (≤30 days) → valid; changing an expiry date requires re-verification

Appendix B — Document & file index

File (in repository `satya928/USMorbit`)	Purpose
`README.md`	Overview, setup, test server steps, stack, KPIs, limitations
`docs/TESTING_GUIDE.md`	Tester accounts, scenarios, how to report issues
`docs/DEMO_SCRIPT.md`	6-minute customer demo walkthrough
`docs/DEPLOYMENT.md`	VPS deployment, upgrades, backups and restore
`docs/FEATURES.md`	Feature checklist and recommended next features
`docs/SECURITY.md`	Security controls, gaps, certification statement
`docs/GOOGLE_SIGN_IN.md`	How to enable Sign in with Google
`docs/USM_Orbit_Beta_Build_Specification.md`	Beta build specification
`docs/USM_Orbit_Project_Plan_and_Status.pdf`	This document
`render.yaml` · `docker-compose.yml` · `Dockerfile`	Hosting configurations
`src/` · `tests/` · `drizzle/`	Application code, automated tests, database migrations

Market figures and competitor developments are taken from the earlier strategy analysis and should be re-verified before use in investor or customer materials.